[INCH-POUND]

MIL-R-12934/35A(EC)
26 November 1990
SUPERSEDING
MIL-R-12934/35(TD)
28 February 1975

MILITARY SPECIFICATION

RESISTOR, VARIABLE, WIRE-WOUND, PRECISION STYLE 2RR2104

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style 2RR2104, variable, wire-wound, precision resistors.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). The PIN shall be of the following form:

RR2104	F	2	A	8	L	203
Style (1.2.2)	Function and shaft length (1.2.3)	Class and grade (1.2.4)	Resistance-temperature characteristic (1.2.5)	Rotational-life characteristic (1.2.6)	Function-conformity-tolerance and resistance-tolerance characteristic (1.2.7)	Resistance (1.2.8)

1.2.2 Style 2RR2104. Style 2RR2104 is a similar style to RR2100.

1.2.3 Function and shaft length. The function and shaft length applicable to this specification shall be identified by symbol F (see figure 1).

1.2.4 Class and grade. The class and grade applicable to this specification shall be symbol 2.

1.2.5 Resistance temperature characteristic. The resistance temperature characteristic applicable to this specification shall be symbol A.

1.2.6 Rotational life characteristic. The rotational life characteristic applicable to this specification shall be symbol 8 for both cups.

1.2.7 Function-conformity tolerance and resistance-tolerance characteristic. The function-conformity tolerance and resistance-tolerance characteristic applicable to this specification shall be L for both cups.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5280 by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905
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1.2.8 Resistance. The nominal total resistance and resolution values applicable to this specification shall be as follows:

First cup - - - - - - - - - 20 kΩ.
Second cup - - - - - - - - - 20 kΩ.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-12934 - Resistors, Variable, Wire-wound, Precision, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-12934, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Mechanical travel. The mechanical travel shall be 3,600° +80°, -0° (+40° at each end) for both cups.

3.4 Effective electrical travel. The effective electrical travel shall be 3,600° +1°, -0° for both cups.

3.5 Function. The type of function applicable to this specification is linear function (independent linearity) for both cups.

4. QUALITY ASSURANCE PROVISIONS

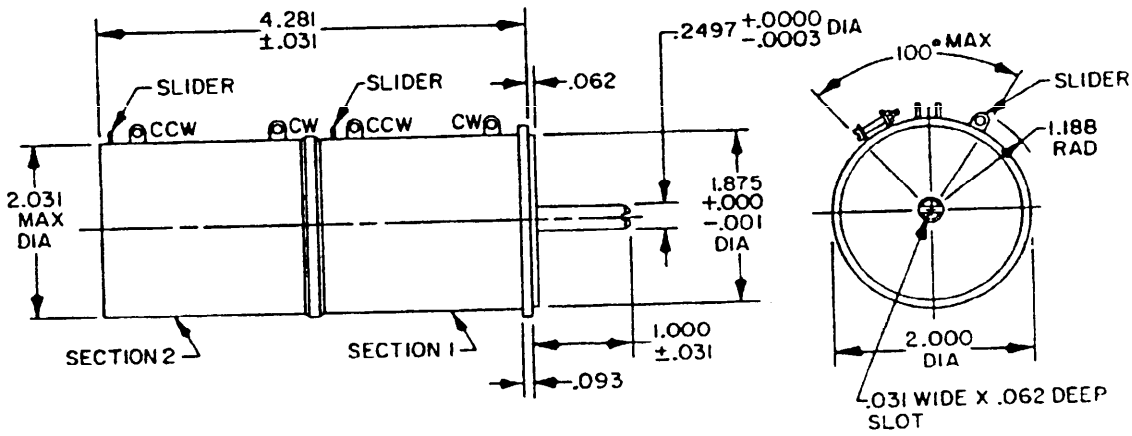
4.1 Inspection. Sampling and inspection shall be in accordance with MIL-R-12934, and as specified herein.

4.2 Verification of qualification. Manufacturers who have passed the qualification requirements, and have been approved and listed on or approved for listing on the qualified products list (QPL) for style RR2100, shall be approved for style 2RR2104.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements shall be in accordance with MIL-R-12934.

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Inches	mm	Inches	mm
.0003	0.008	1.000	25.40
.001	0.03	1.188	30.18
.031	0.79	1.875	47.63
.062	1.57	2.000	50.80
.093	2.36	2.031	51.59
.2497	6.342	4.281	108.74

Style	Turns	Maximum continuous working voltage	Power rating (watts)	Maximum starting and running torque				Stop torque (inch-pound)
				First cup		Second cup		
				starting	running	starting	running	
2RR2104	10	500	5 1/	2.0	1.0	2.0	1.0	35

1/ When ten-turn units are gauged, all cups shall be 75° of rated wattage.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Style 2RR2104.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-12934 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Navy - EC

Preparing activity:
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Agent:
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(Project 5905-N310)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-12934/35A(EC)	90/11/26

3. DOCUMENT TITLE
Resistor,Variable,Wire-Wound,Precision,Style 2RR2104

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
NAVY-EC	(1) Commercial 513-296-5257	(2) AUTOVON 986-5257
c. ADDRESS (Include Zip Code) DESC-ECT 1507 Wilmington Pike Dayton, OH 45444-5280	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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